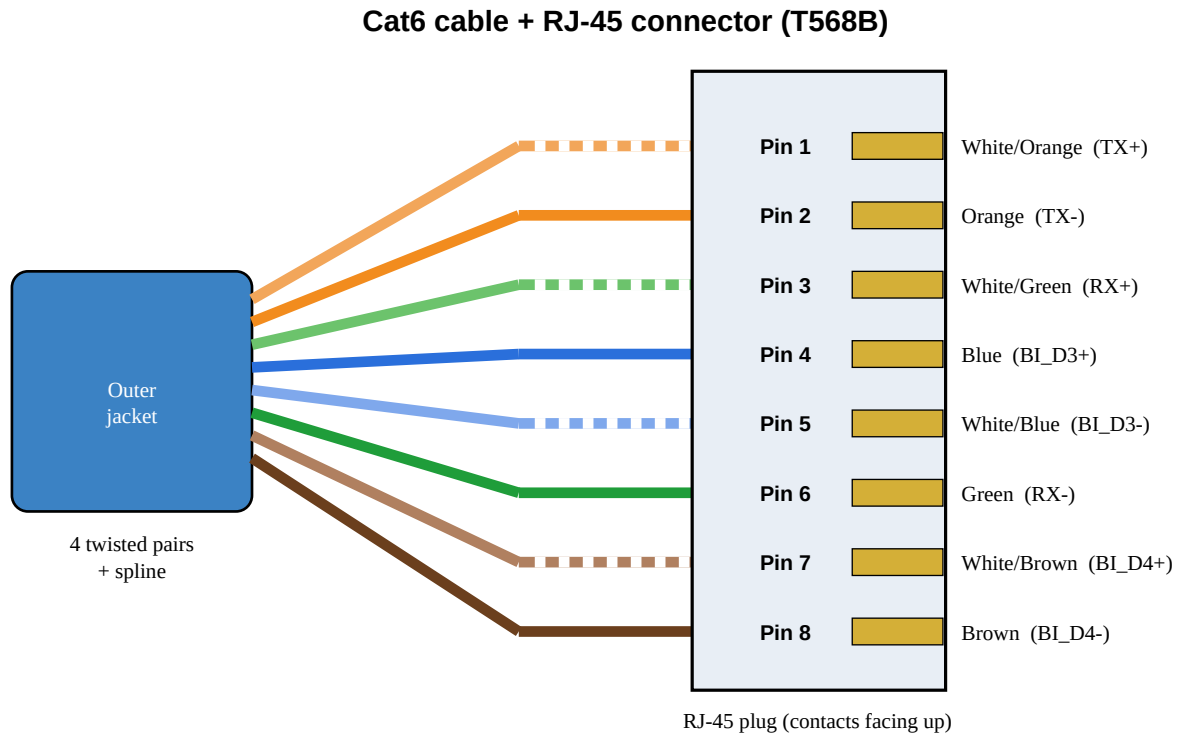


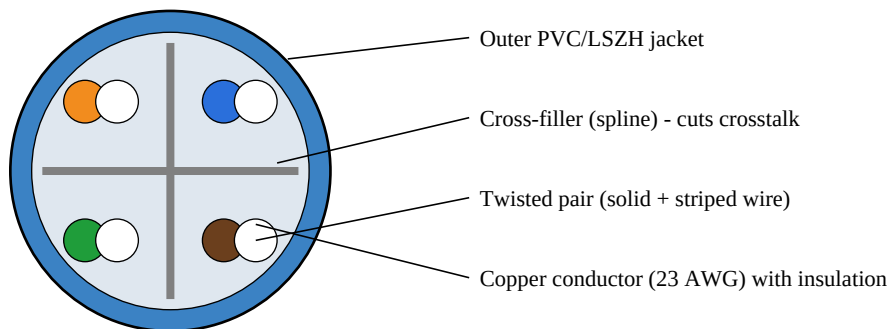
Network Cables and Connectors

Assignment Answers

1. Labeled diagram: Cat6 cable, T568B colour order, RJ-45 pins



Cross-section of a Cat6 cable



Pin	Colour (T568B)	Signal (Gigabit)
1	White/Orange	TX+
2	Orange	TX-
3	White/Green	RX+
4	Blue	BI_D3+
5	White/Blue	BI_D3-
6	Green	RX-
7	White/Brown	BI_D4+
8	Brown	BI_D4-

Tip: Hold the plug with the clip facing down and the contacts facing you; pin 1 is on the left. Order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.

2. Three types of network cables and where they are used

Cable	Real-world use	Why
Cat5e (copper, up to 1 Gbps, 100 m)	Home/office WiFi router to laptop or TV; Zomato POS terminal at a restaurant counter	Cheap, easy to crimp, and 1 Gbps is more than enough for billing data and normal internet speeds.
Cat6 / Cat6a (copper, 1-10 Gbps, shielded options)	Connecting gaming consoles/PCs and screens to a switch at an IPL screening / gaming zone ; office servers	Thicker with a spline separator, so less crosstalk and stable high speed with low lag for live video and gaming.
Fiber optic (glass, light signals, 10-100+ Gbps)	Backbone link to an IPL streaming server / data centre; ISP connection to the building	Huge bandwidth over long distances (km), no electrical interference, and very low latency for millions of viewers.

3. How to crimp an RJ-45 connector (step by step)

Note: I could not watch the YouTube video myself, so this is written from standard crimping practice. Please compare it with the video you watch.

- 1 **Gather tools:** Ethernet cable, RJ-45 plug, crimping tool with cutter/stripper, cable tester.
- 2 **Strip the jacket:** Remove about 2.5 cm (1 inch) of the outer jacket without nicking the inner wires.
- 3 **Untwist and straighten:** Separate the four pairs and straighten each wire. Cut off the plastic spline.
- 4 **Arrange in T568B order:** White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
- 5 **Trim evenly:** Cut the wires flat, leaving about 1.2 cm (1/2 inch) sticking out from the jacket.
- 6 **Insert into the plug:** Hold the clip side down and slide the wires in until each reaches the end of the plug. The jacket should also enter slightly.
- 7 **Crimp:** Place the plug in the crimping tool and squeeze firmly so the gold pins pierce the wires.
- 8 **Test:** Repeat on the other end, then check with a cable tester. All 8 lights should glow in order.

Two common mistakes to avoid

- **Wrong colour order or untwisting too much:** A wrong order causes no link or slow speed, and untwisting more than 1.3 cm increases crosstalk.
- **Wires not pushed fully to the end / cutting them uneven:** Some pins will not touch the wires, giving a loose or dead connection.

4. Gaming zone for an IPL screening event

Goal: Connect multiple screens and consoles with high speed, low lag and reliability.

Connection	Cable and connector	Reason
Consoles/PCs to network switch	Cat6 (Cat6a if budget allows) patch cables with RJ-45 connectors	Supports 1 Gbps (10 Gbps on short runs), low latency, good shielding from interference in a crowded room.
Console/media player to screens/projector	HDMI 2.1 cable (certified Ultra High Speed), HDMI connector	Carries 4K 120 Hz video and audio, which is what the screens need for smooth live matches and gaming.
Switch to switch or to the main router/internet (long runs)	Fiber optic (multimode OM3/OM4) with LC connectors	Fast backbone for many users, immune to electrical noise from lights and speakers, works over long distances.
Streaming box / live feed device	Cat6 wired Ethernet instead of WiFi	Wired links avoid buffering and dropped frames when many devices share the WiFi.

Extra tips

- Use a **Gigabit managed switch** (with enough ports plus a few spare) and keep each copper run under 100 m.
- Use factory-made, tested patch cables and label both ends of every cable.
- Route cables away from power cords, and use cable covers/trays for safety.
- Keep a few spare cables and a cable tester ready for the event.

5. Photo of a network cable or connector in use

I cannot take or find a photo of your own setup, so please add your photo in the box below. A sample description is given that you can edit to match your photo.

[Paste your own photo here]

Sample description (example):

- **Where:** Behind a home WiFi router.
- **Type:** A blue Cat5e/Cat6 Ethernet patch cable (UTP) with clear plastic RJ-45 connectors.
- **What it connects:** One end is plugged into a LAN port on the router and the other into a smart TV (or a desktop PC). The router's WAN port is connected to the ISP's fiber ONT by a separate cable.
- **Why it is used:** A wired link gives a steadier, faster connection than WiFi for streaming or gaming.